

Modeling of Interspecific Interaction of *Yersinia pseudotuberculosis* and *Listeria monocytogenes* in a Mixed-Species Biofilms with Microorganisms Isolated from Coastal Waters of the Sea of Japan

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We studied the possibility of formation of mixed-species biofilms by *Yersinia pseudotuberculosis* and *Listeria monocytogenes* with marine saprotrophic bacteria *Flavobacterium* sp. and *Micrococcus luteus* isolated from the coastal waters of the Sea of Japan in summer. Model experiments showed that *Flavobacterium* sp. and *Micrococcus luteus* can form both single- and mixed-species biofilms with the specified pathogenic bacteria thus stimulating their growth. This can contribute to the preservation of the pathogens in the marine environment.

Key Words: *mixed-species biofilm; Yersinia pseudotuberculosis; Listeria monocytogenes; marine saprotrophic bacteria; interspecific interaction*

Biofilm is a structurally and functionally organized community of microorganisms, in which bacteria form a common genetic system [2]. Biofilms formed by a single species of bacteria have been the subject of many studies [5,15]. However, it is now clear that *in vivo* biofilms are more often formed by not one, but several species [13]. Due to this, of great fundamental and practical importance is the study of mixed-species biofilms, rather than experimental studies of structures formed by one type of microorganism [9].

Yersinia pseudotuberculosis and *Listeria monocytogenes* are pathogens inducing sapronoses. Depending on the environment, they can be parasitic (when ingested by a warm-blooded organism) and saprophytic (living in the environment) [8]. In the environment, these bacteria are common on plants, in soil and water. They can be introduced into marine environment by flood and groundwater or water pollution

from sewage, which pose a potential danger for the population. For example, bivalve mollusks infected with *L. monocytogenes* regularly cause outbreaks of listeriosis worldwide [10,15].

Marine environment is not favorable for pathogenic bacteria; therefore, the formation of biofilms is a mechanism of their adaptation aimed at the maintenance of their population under these conditions [2]. In light of this, it is important to find out to what extent the marine environment is a “cemetery” or a “springboard” for the formation of epidemic variants of pathogenic bacteria [10]. In particular, experimental and epidemiological assessment of the possible consequences of interactions between marine microorganisms, permanent inhabitants of the oceanic ecosystem, and microorganisms introduced with waste water is required.

Our aim was to assess in model experiments the ability of saprotrophic bacteria isolated from the marine environment to form biofilms with *Y. pseudotuberculosis* and *L. monocytogenes*, as well as the possibility of interaction of marine microorganisms with introduced bacteria in the biofilm.

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MATERIALS AND METHODS

Coastal waters of the Sea of Japan (Nakhodka Bay significantly affected by industrial and domestic wastewater and Vostok Bay burdened with the entry of organic pollutants) [1] were chosen for the study. Water samples were taken in summer at 20°C. During microbiological studies, 35 strains of saprotrophic marine microorganisms were isolated: 17 from Nakhodka Bay and 18 from Vostok Bay. Identification of the obtained isolates was carried out by their morphological, cultural, and biochemical properties. To determine the genera of the studied bacteria, differentiating tests were carried out in accordance with the Bergey's Manual of Systematic Bacteriology [11]. The taxonomic status of the strains was confirmed using strips with biochemical tests API 20 (BioMerieux). The results were interpreted using APIWEB database (<https://apiweb.biomerieux.com/>).

For the formation of mixed biofilms, we used *L. monocytogenes* strain (museum inventory No. 870) and *Y. pseudotuberculosis* strain (museum inventory No. 3515) from the collection of G. P. Somov Research Institute of Epidemiology and Bacteriology.

The film-forming capacity of all the studied bacteria was assessed in 24-well plates. Culturing in a concentration of 10^3 CFU/ml was carried out in a liquid medium for marine microorganisms (MMM) [6] at 20°C for 3 days.

Single- and mixed-species biofilms were detected on an abiotic surface using the spectrophotometric method [14]. Optical density (OD) was measured over 3 days of incubation in 96-well plates on a Labsystems iEMS Reader MF (Bio-Rad) at $\lambda=540$ nm. Wells containing medium without bacteria served as the control. By analogy with the study [12], all the studied microorganisms by OD values were divided into two groups: with low ($OD < 0.4$) and high ($OD > 0.4$) biofilm-forming capacity.

To determine the dynamics of bacterial reproduction in monoculture and in the association, the number of microorganisms in the planktonic and biofilm phases was determined. The growth of biofilms formed by each strain was studied on Petri dishes. The number of bacteria was expressed in CFU/ml.

Culturing of marine bacteria was carried out on a dense nutrient medium for quantification of saprophytic heterotrophic bacteria in marine environment (MMM) [6]; *L. monocytogenes* were cultured in DDSL medium (differential diagnostic environment for *Listeria*) [7] and *Y. pseudotuberculosis* were cultured in medium 67 (Serov) at room temperature (20°C).

The graphs show $M \pm m$ for triplicates for each specific variant of the experiment. Statistical processing of the results was performed using Microsoft Excel

2007. Significance was assessed by the Student's *t* test; the differences were significant at $p \leq 0.05$.

RESULTS

High ability of *L. monocytogenes* and *Y. pseudotuberculosis* to form single-species biofilms was demonstrated in previous studies [3,9]. Here we found that all strains of marine microorganisms used in the work also formed biofilms. Of 35 marine bacteria used in the present study, high biofilm-forming properties were detected in *Micrococcus luteus* (isolated from Nakhodka Bay, serial No. 3), and *Flavobacterium* sp. (isolated from Vostok Bay, serial No. 9). These marine microorganisms were used in a model experiment on the formation of two-species biofilms. The experiment was performed in 70 variants: for studying formation of biofilm by saprotrophic microorganisms from Nakhodka Bay with pathogens (34 variants) and saprotrophic microorganisms from Vostok Bay with pathogens (36 variants).

The experiments showed that the highest OD during co-cultivation were achieved in only 2 pairs of biofilms: *M. luteus* and *Y. pseudotuberculosis* 3515; *Flavobacterium* sp. and *L. monocytogenes* 870 ($M \pm \sigma$: 0.87 ± 0.02 and 0.92 ± 0.04 opt. density units, respectively). In other multi-species biofilms, OD varied from 0.05 ± 0.03 to 0.43 ± 0.09 ($p < 0.05$). Therefore, these pairs were used for evaluation of interspecific interactions in mixed-species biofilms.

When studying the dynamics of reproduction, we found that all the studied bacteria form a mature single-species biofilm on the second day. However, in the mixed biofilm, the number of marine bacteria of both strains on the first day of culturing with pathogenic microorganisms was lower than in the single-species biofilm. In mixed biofilm of *Yersinia* sp. and *Micrococcus* sp., the number of *Yersinia* sp. on day 2 was higher by 3 orders of magnitude (Fig. 1, *b, d*). At the same time, the number of *Micrococcus* sp. in the single-species biofilms was significantly higher than after cultivation with *Yersinia* sp. ($p < 0.05$; Fig. 1, *a, b*). During the next days of the experiment, yersinia more actively formed mixed biofilm and proliferated than micrococci and then yersinia in single-species biofilm (Fig. 1, *c, d*). The growth of yersinia in single-species biofilm slowed down by the third day of the experiment, while in mixed biofilm, its growth rate significantly increased from the first to the third day.

Similar changes were recorded when assessing the dynamics of biofilm formation by marine microorganisms with listeria throughout the observation period (Fig. 2). In the mixed biofilm formed by *Listeria* sp. and *Flavobacteria* sp., pathogenic bacteria prevailed

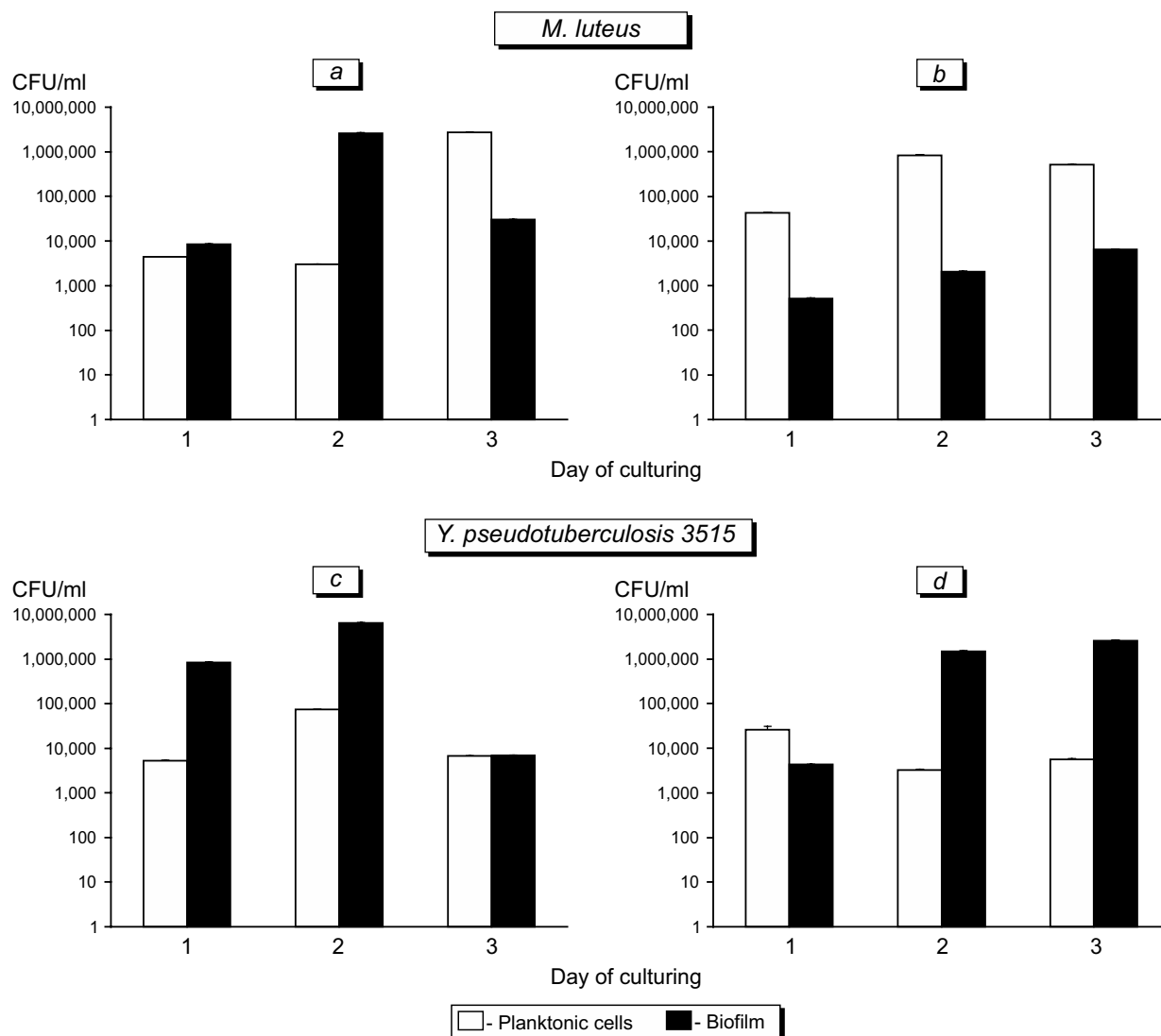


Fig. 1. Dynamics of the number of *Y. pseudotuberculosis* and *M. luteus* in monoculture (a, c) and in a mixed biofilm (b, d).

over the saprotrophic marine microorganisms (Fig. 2, b, d). *Listeria* from mixed biofilm proliferated more actively on the second day of culturing (Fig. 2, b), but their number was by 2 orders of magnitude higher than in the single-species biofilm (Fig. 2, a). *Flavobacteria* in both monoculture and in mixed biofilm showed almost the same growth rates (Fig. 2, a, b).

The presented results suggest that in both analyzed cases, the growth of pathogenic bacteria is stimulated by saprotrophic marine microorganisms. There are data on the interaction of *Listeria* with saprotrophic microorganisms isolated from the soil [4] and that metabolites released by saprotrophic bacteria can stimulate the growth of *L. monocytogenes* [4]. If we compare these results with those obtained by us, we can assume that metabolites activated the growth of *L. monocytogenes* and *Y. pseudotuberculosis*.

Thus, the conducted model experiments showed

that *Flavobacterium* sp. and *M. luteus* can form single-species and mixed biofilms with the specified pathogenic introduced microorganisms and stimulate their growth. This can affect both the persistence of *Y. pseudotuberculosis* and *L. monocytogenes* in the marine environment and their pathogenic potential. As the experiments were carried out under natural conditions, the results obtained suggest that similar processes may occur in bacterial communities in the coastal waters of the Sea of Japan.

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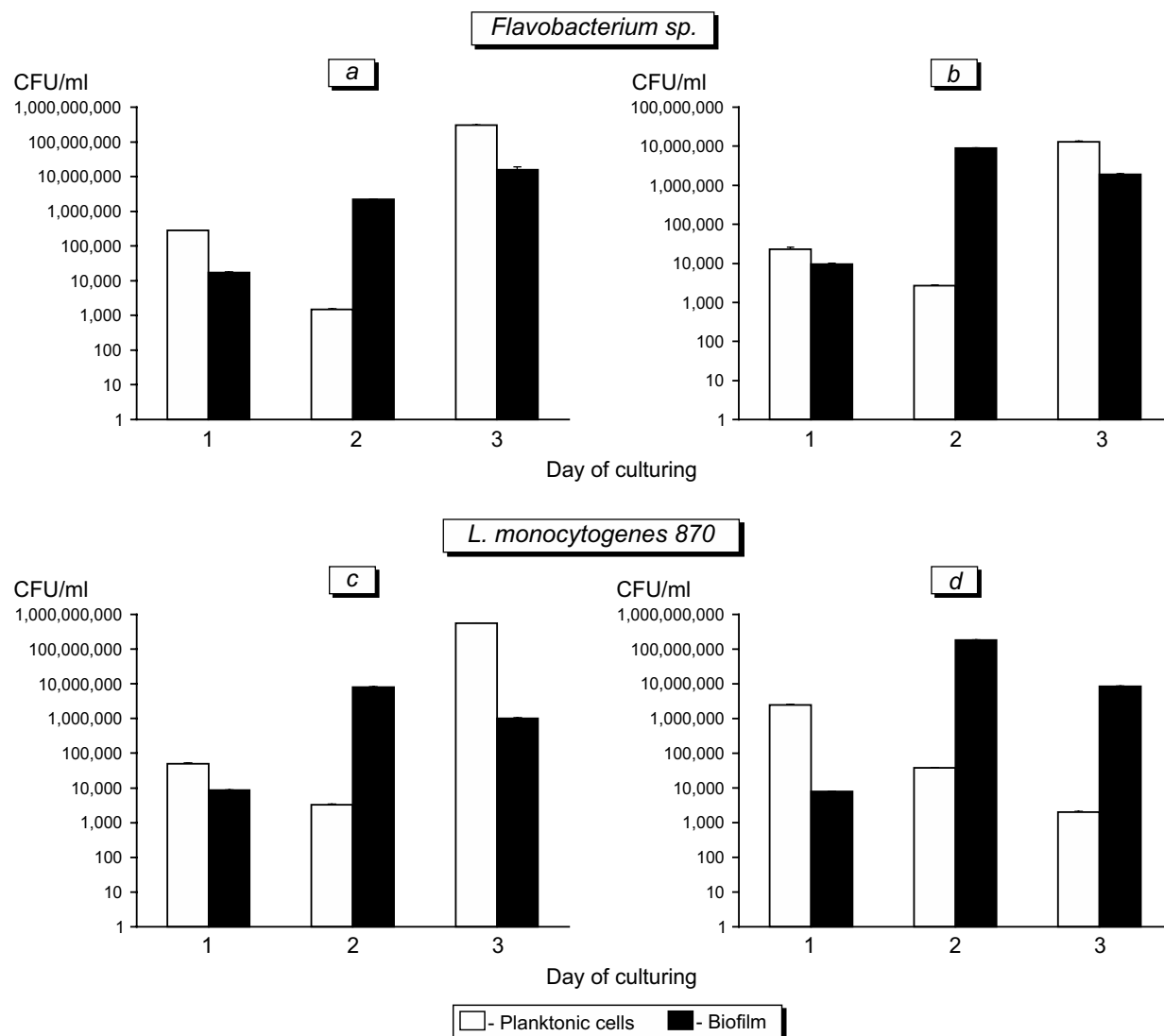


Fig. 2. Dynamics of the number of *L. monocytogenes* and *Flavobacterium* sp. in monoculture (a, c) and in a mixed biofilm (b, d).

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